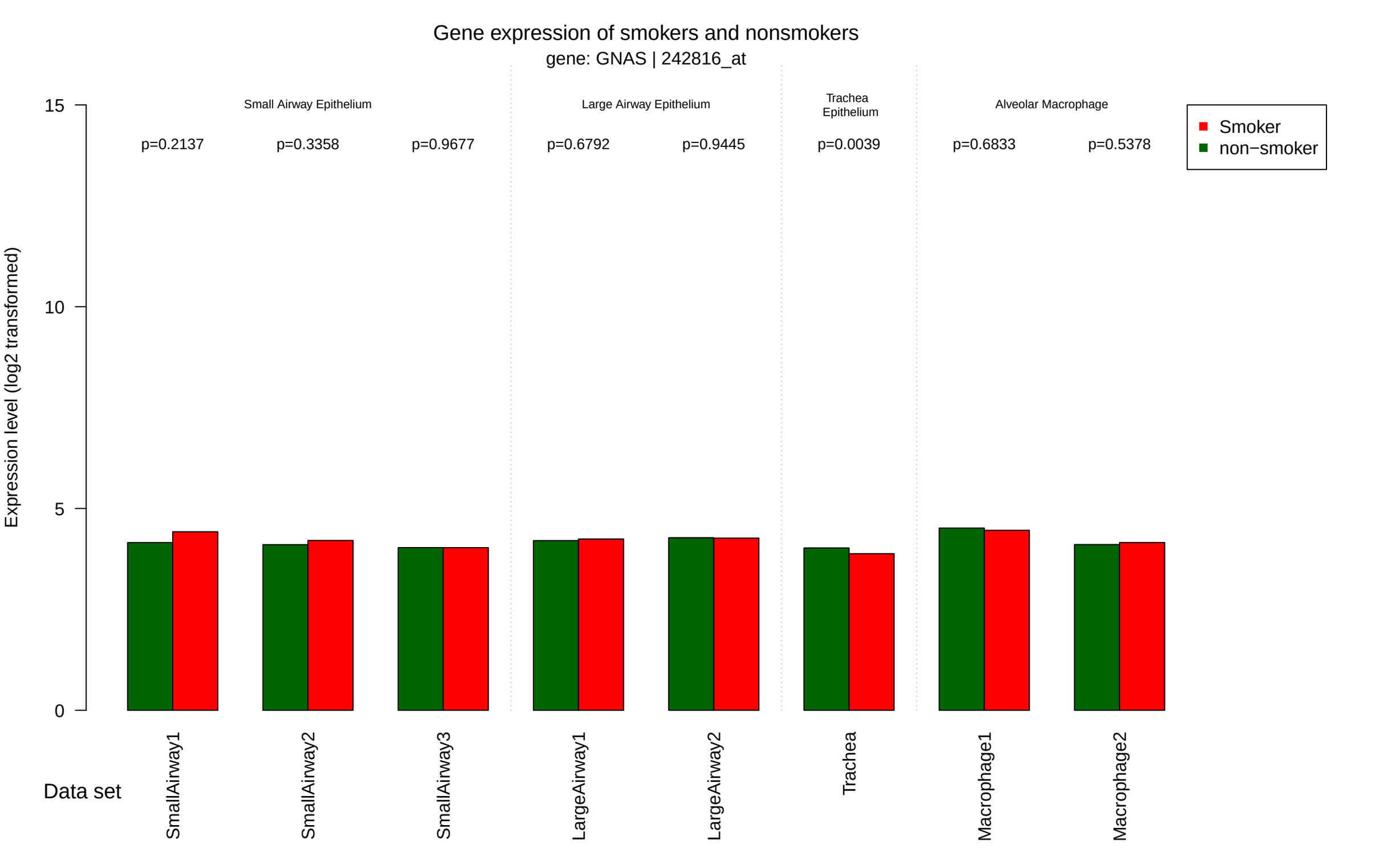
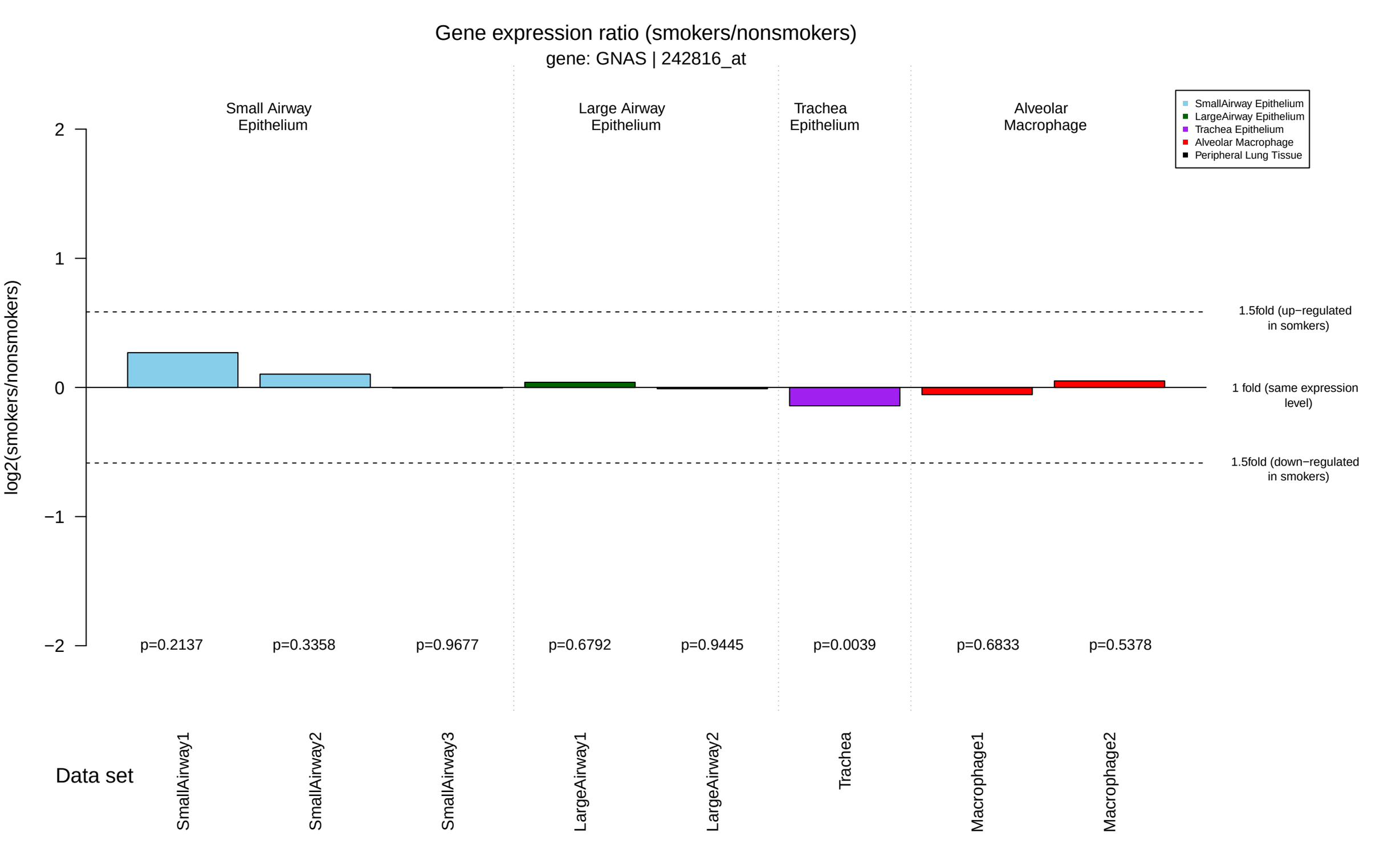
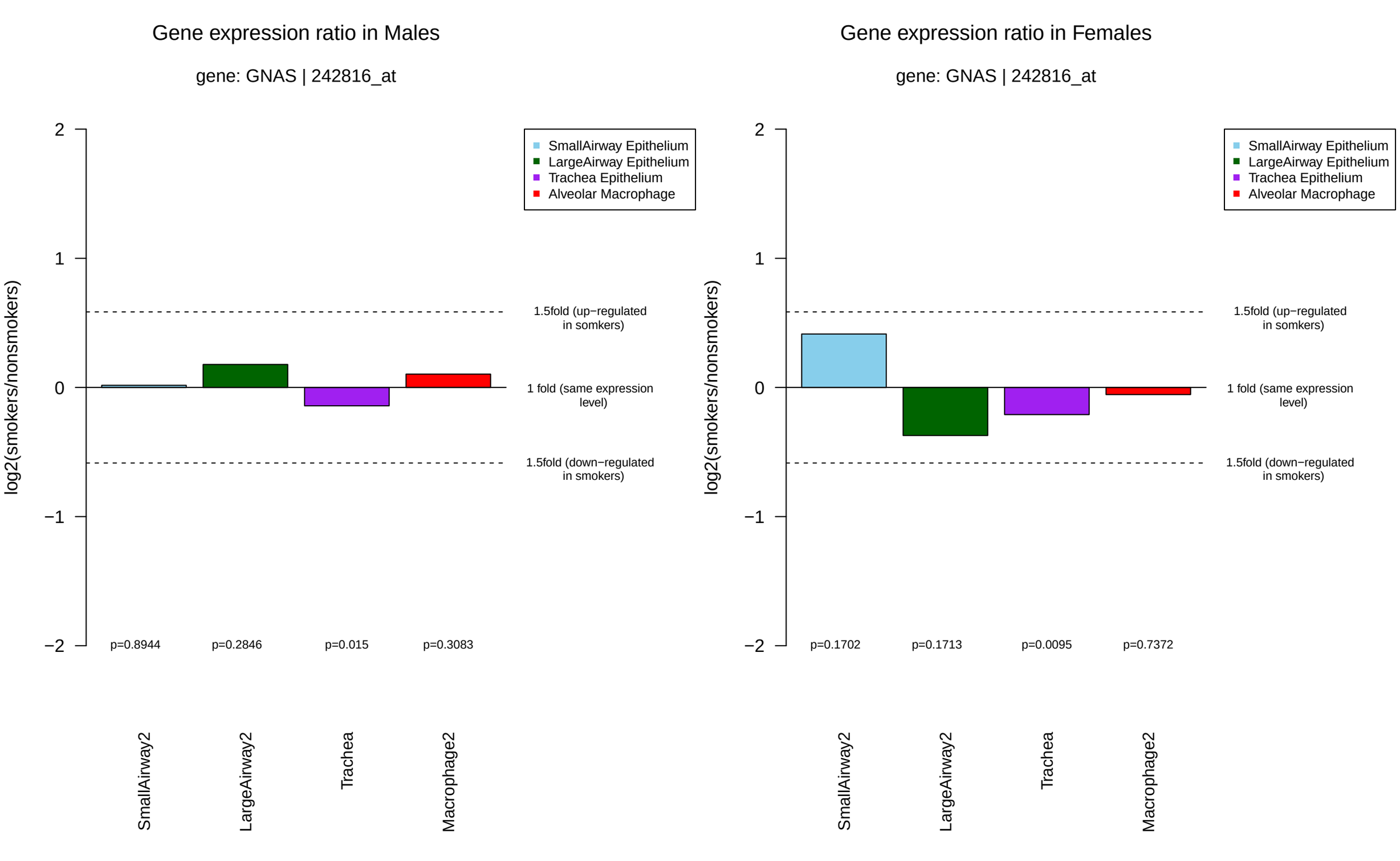
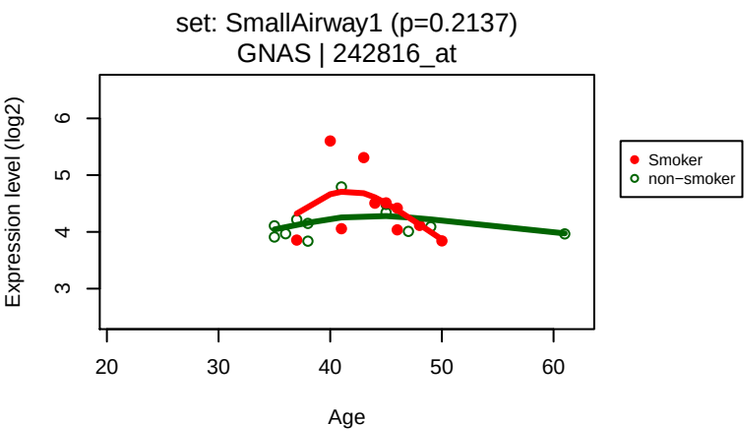
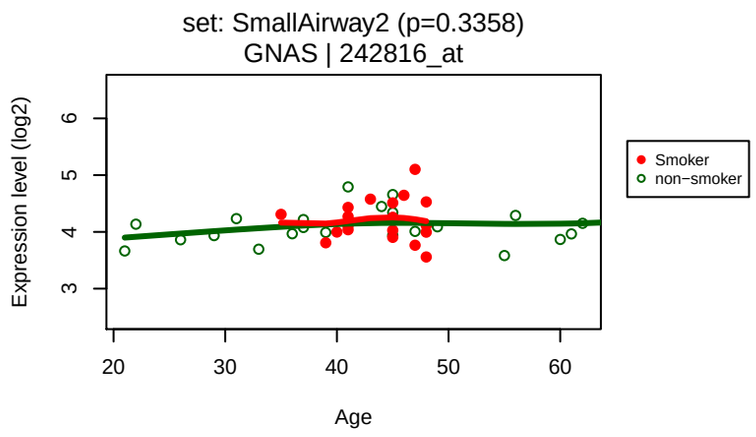
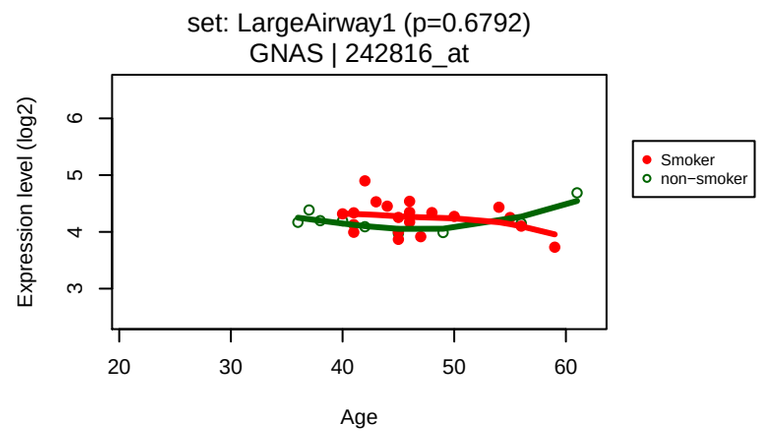

Gene expression of smokers and nonsmokers
gene: GNAS | 242816_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2137
p=0.3358
p=0.9677
p=0.6792
p=0.9445
p=0.0039
p=0.6833
p=0.5378
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

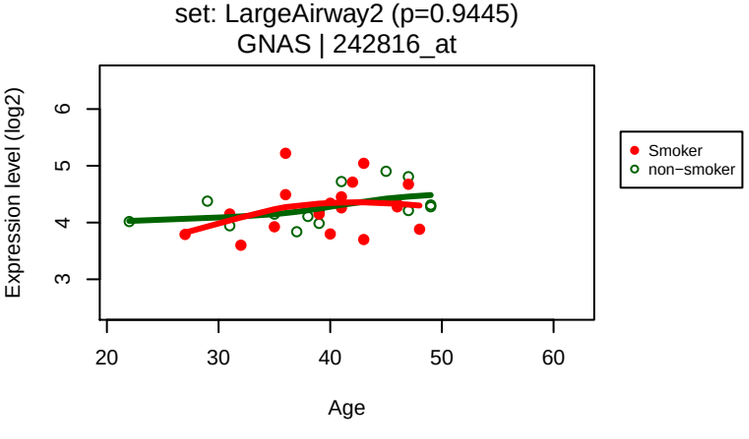

Gene expression ratio (smokers/nonsmokers)
gene: GNAS | 242816_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2137
p=0.3358
p=0.9677
p=0.6792
p=0.9445
p=0.0039
p=0.6833
p=0.5378
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

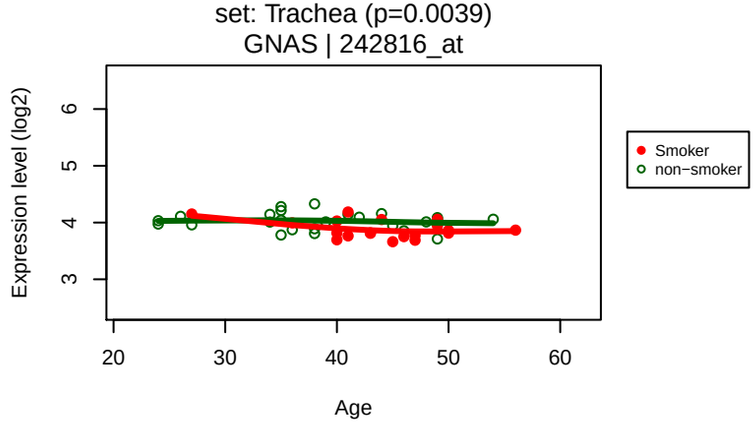

Gene expression ratio in Males
gene: GNAS | 242816_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8944
p=0.2846
p=0.015
p=0.3083
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNAS | 242816_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1702
p=0.1713
p=0.0095
p=0.7372
SmallAirway2
LargeAirway2
Trachea
Macrophage2

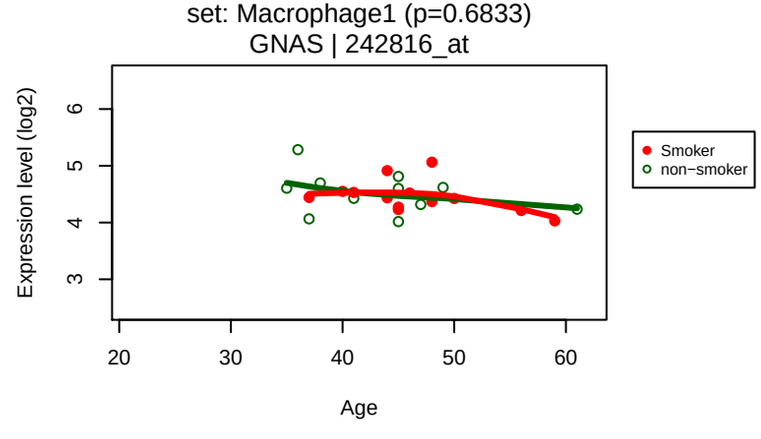

set: SmallAirway1 (p=0.2137)
GNAS | 242816_at
Expression level (log2)
Age
Smoker
non−smoker

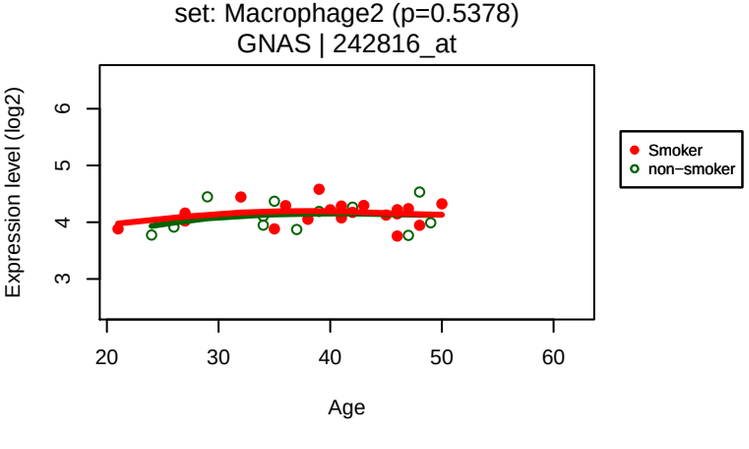

set: SmallAirway2 (p=0.3358)
GNAS | 242816_at
Expression level (log2)
Age
Smoker
non−smoker

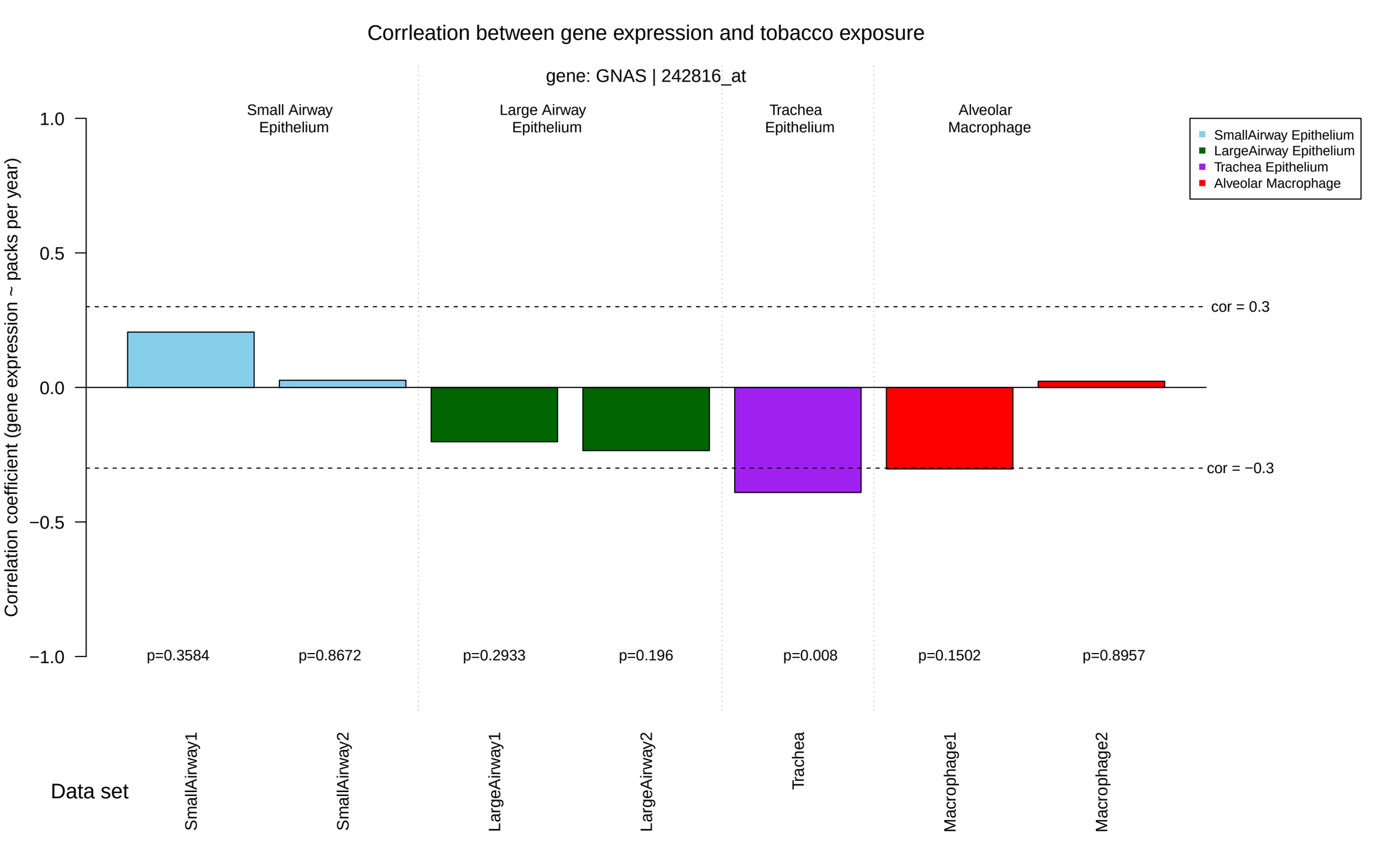

set: LargeAirway1 (p=0.6792)
GNAS | 242816_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9445)
GNAS | 242816_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0039)
GNAS | 242816_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6833)
GNAS | 242816_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5378)
GNAS | 242816_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNAS | 242816_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3584
p=0.8672
p=0.2933
p=0.196
p=0.008
p=0.1502
p=0.8957
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2